

KNN

K-Nearest Neighbors

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Introduction to KNN

Tell me about your friends and I will tell you who you are

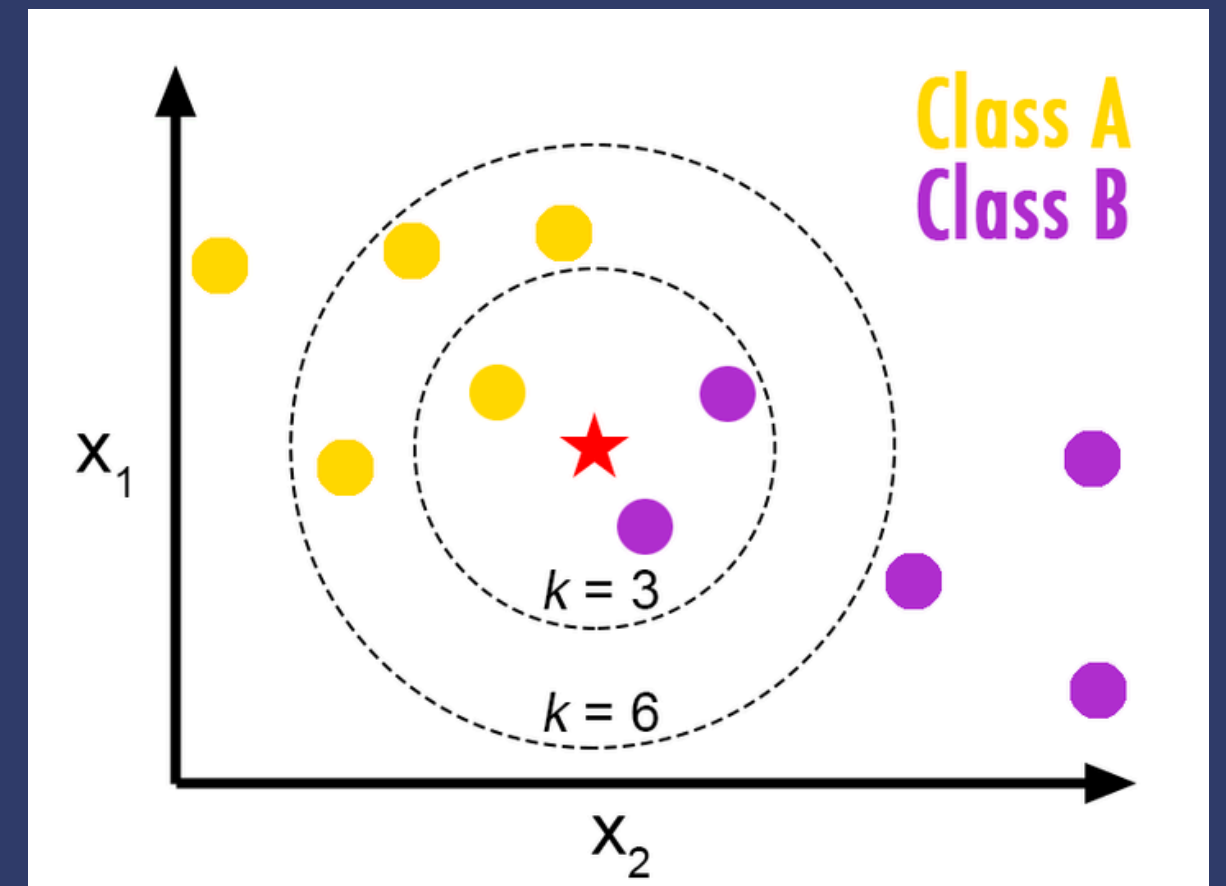


- KNN is simple supervised machine learning algorithm.
- The main idea is that we make predictions based on the closest “neighbors” to a new data point.
- It’s widely used in both classification and regression tasks and is very popular because it’s easy to understand, easy to implement

What is the K-Nearest Neighbors algorithm?



- The KNN algorithm works by comparing a new data point to the data we already have.
- When we want to predict the label of a new point, the algorithm finds the K closest points to it



Why is KNN considered a
lazy learning / instance-based
algorithm?

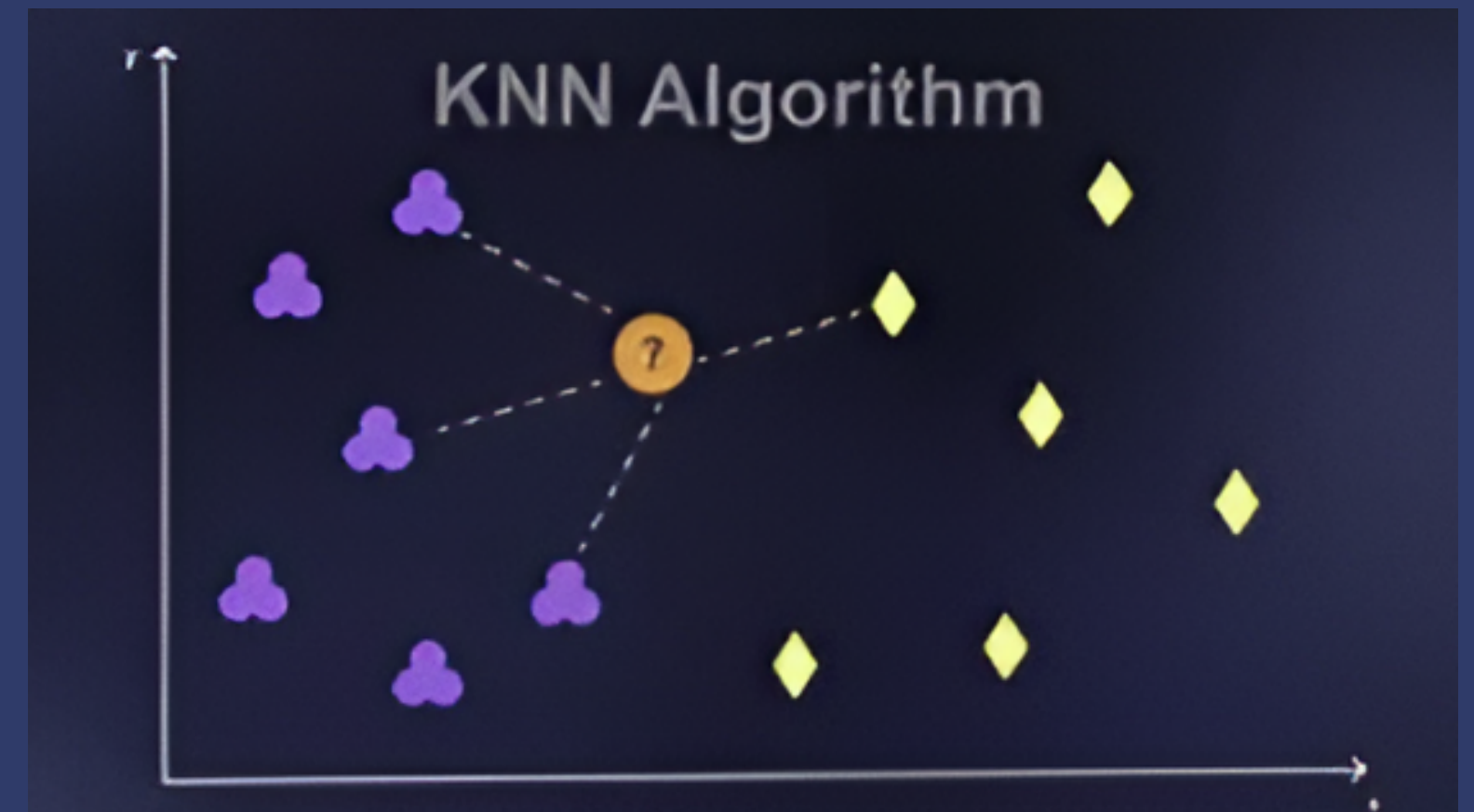


- There is no actual training phase the algorithm simply stores all the training data.
- The “work” happens only during prediction, when it searches for the nearest neighbors.
- Training Time → Fast
- Prediction Time → Slow
- It's also known as an instance-based algorithm because it relies directly on the data instances themselves, not on a pre-learned model

How KNN Works

The Idea of Distance Between Data Points

animation



Step 1 → **Calculate the distance**

The algorithm starts by calculating the distance between the new data point and every single point in the dataset.

Step 2 → **Sort the distances**

After computing all the distances, KNN sorts the points from the closest to the farthest based on the distance values.

Step 3 → **Select the nearest K neighbors**

Once the points are sorted, the algorithm takes the top K closest points — these are the neighbors that will influence the prediction.

Step 4 → **Make the prediction**

Classification → Mode

Regression → Mean

Distance metrics

Euclidean Distance

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

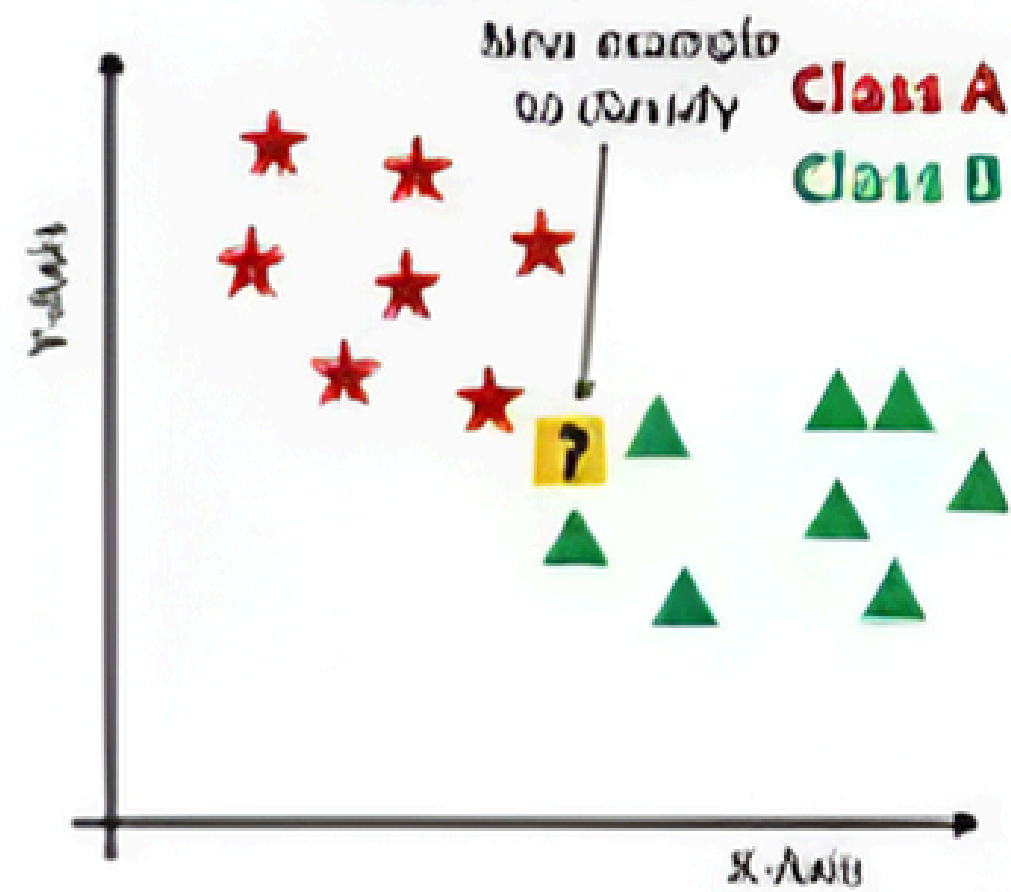
Manhattan Distance

$$\text{Mdist} = |x_2 - x_1| + |y_2 - y_1|$$

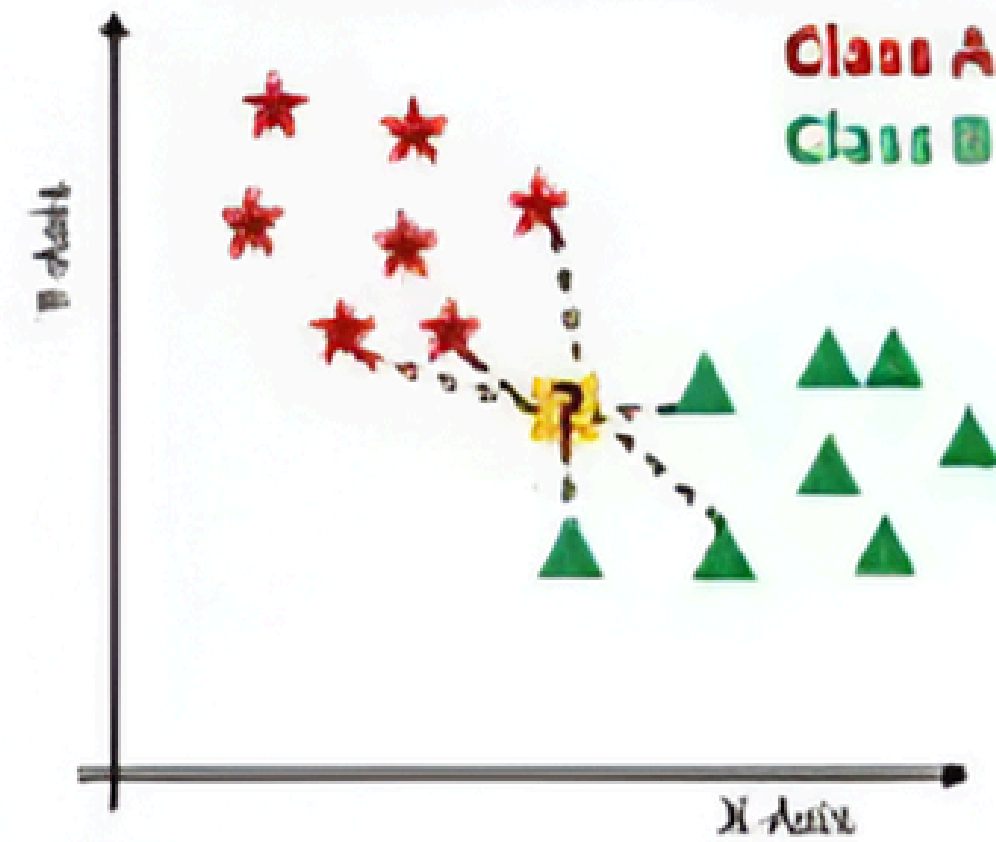
Minkowski Distance

$$D(x, y) = \left(\sum_{i=1}^n |x_i - y_i|^p \right)^{1/p}$$

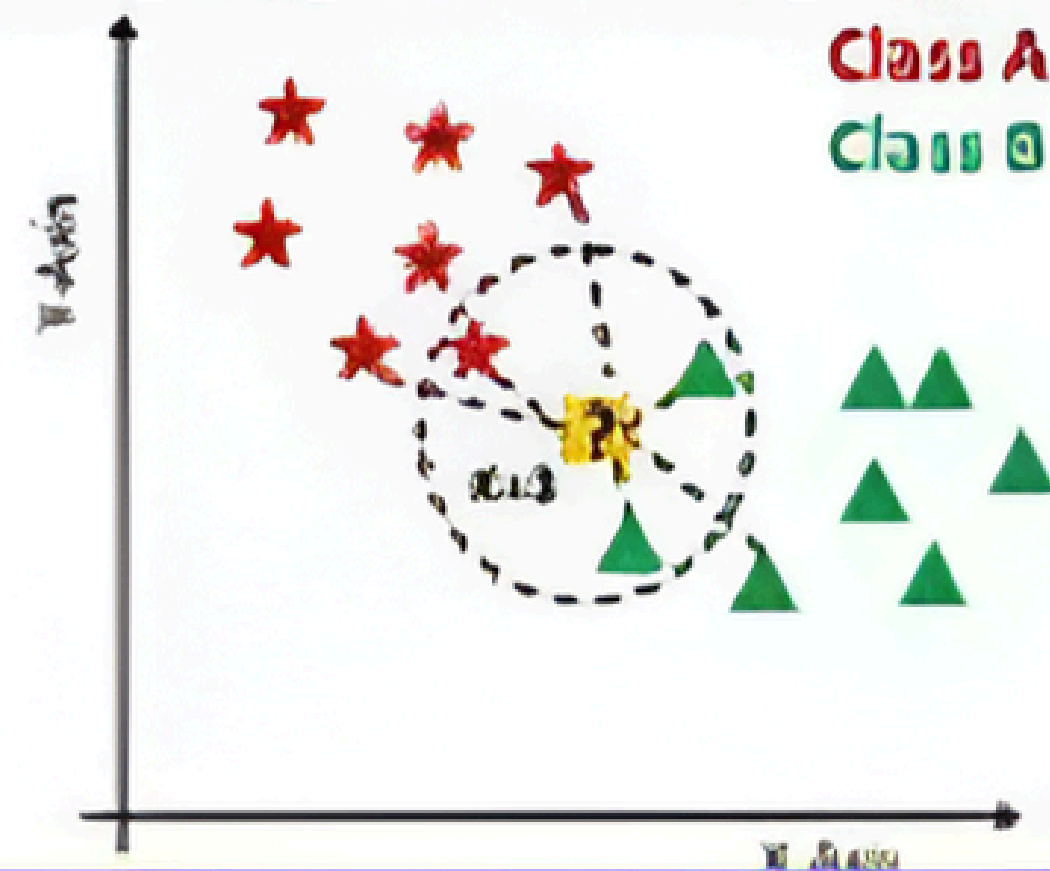
Initial Data



Calculate Distance



Finding Neighbors & Voting for Labels



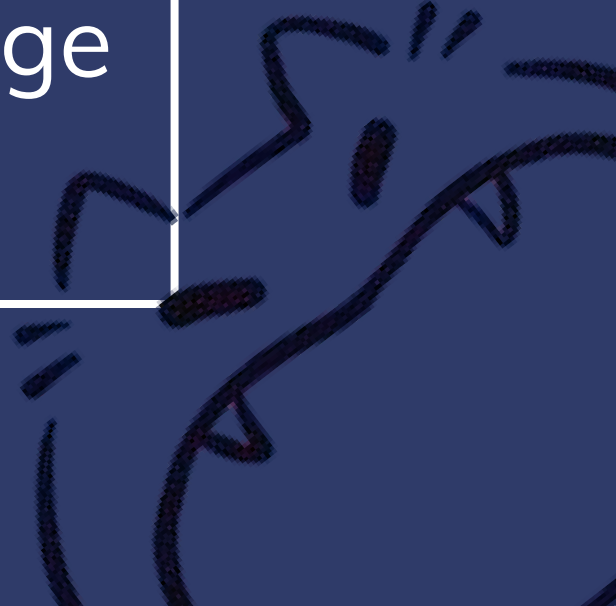


The Prediction Process: Classification vs Regression





Aspect	Classification	Regression
Output Type	Class label	Numerical value
Prediction Method	Majority vote	Average/Mean
Example Use	Spam detection	Price prediction
Distance Impact	Same	Weighted average often better



How to Choose Value of K

Small vs Large





K-Nearest Noodles



من جارك القريب ولا أخوك البعيد



Customers Data

Asian Food

Tolerance

Age

Budget

Eat Out / Month

Dragon Spicy
Noodles



Small $K = 1$

New Customer

Age : 25
Tolerance : 7
Asian Food : 2
Eat out / month : 6
Budget : 25\$

Nearest Customer Loves
Dragon Spicy Noodles ?

NO!!!

He had a stomach upset

Is that An Outlier ??

Nearest Customer

**Age : 26
Tolerance : 9
Asian Food : 3
Eat out / month : 9
Budget : 24\$**



Large K = 50

New Customer

Age : 25

Tolerance : 7

Asian Food : 2

Eat out / month : 6

Budget : 25\$

Customer 1

It is TOO SPICY,
HOW AM I SUPPOSED
TO EAT THAT THING ?

Customer 3

PAID \$18 for what felt
like CHEWING ON
LAVA.

Customer 2

My soul left my body
after ONE BITE and
HASN'T RETURNED.

Customer 4

This isn't food, it's a
chemical weapon
disguised as noodles.

???

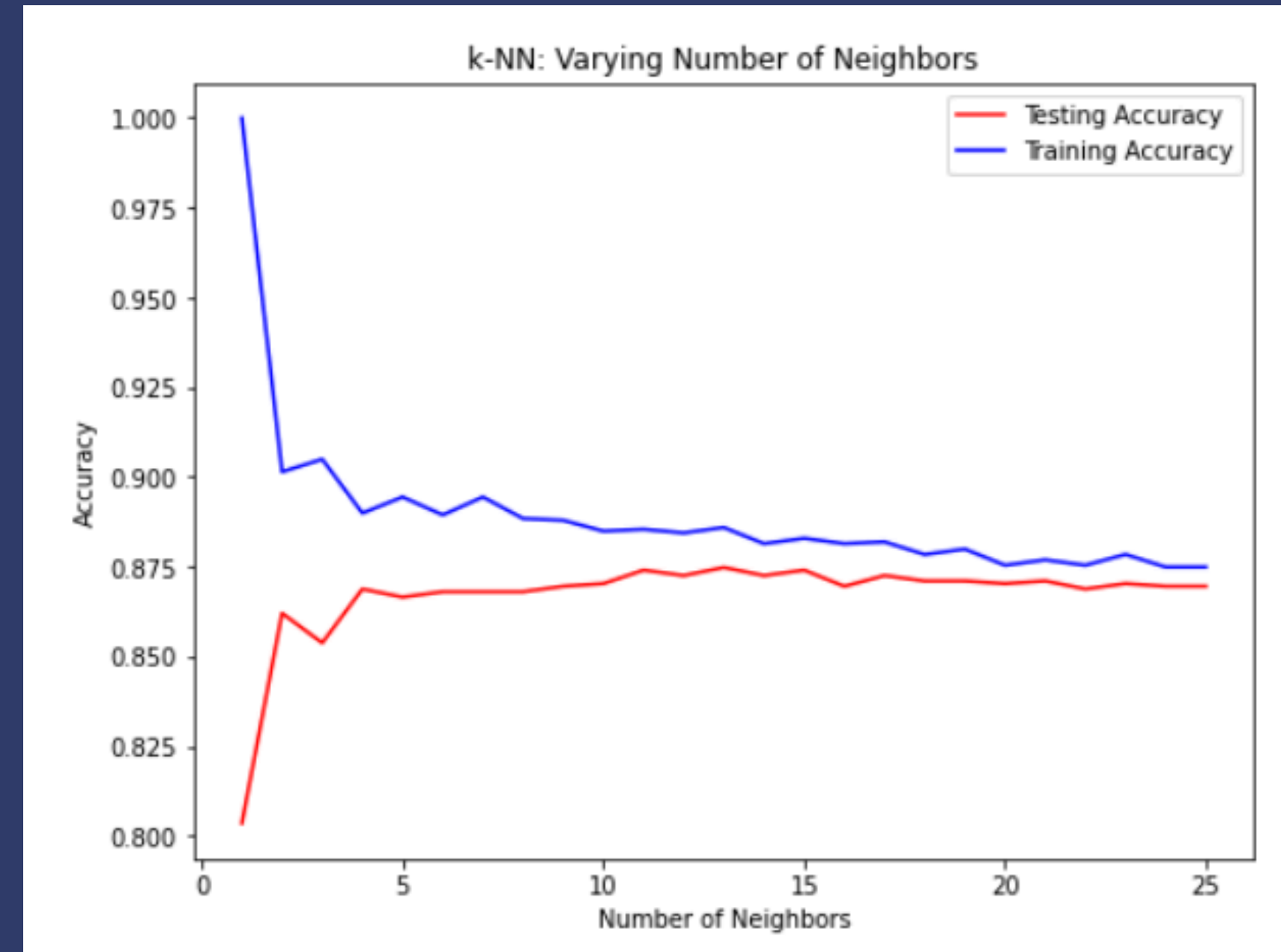
70 % NO

30 % YES

.....

Various Values of K

- **small k**
 - Sensitive to Outliers
 - High variance , Low Bias
 - Complex Boundary
- **large k**
 - Smooth prediction
 - Low variance , High Bias
 - Simple Boundary





How to Choose Value of K

- $k = 5$ (default)
 - Test Multiple k using Cross Validation
 - Rule of thumb : $k = \text{root}(n)$
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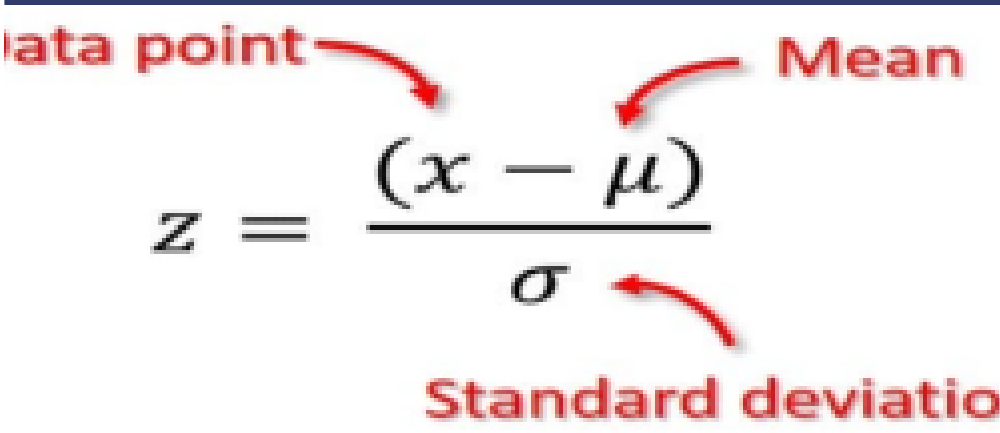
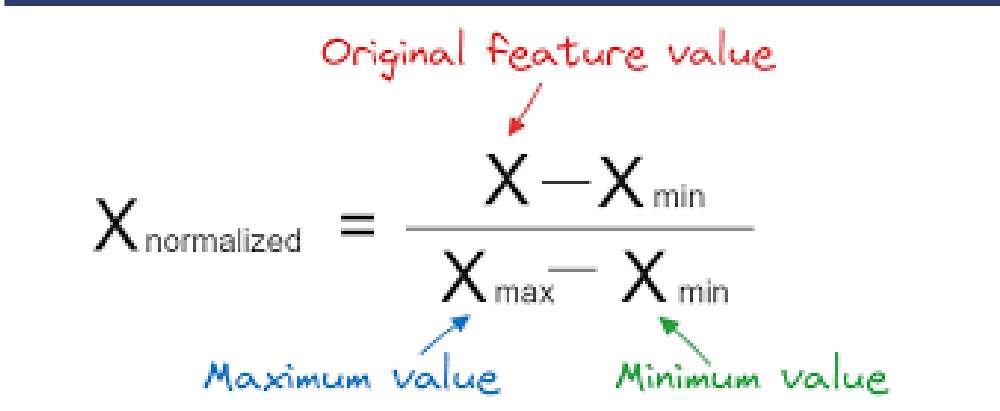


Feature Scaling – The Mathematical Prerequisite



Why Scaling is Necessary for KNN

- KNN is a distance-based algorithm (e.g., using Euclidean distance), meaning its predictions are based entirely on geometric proximity in the feature space
- If features have widely different numerical scales (e.g., age vs. income), the feature with the largest magnitude will dominate the distance calculation .
- This Feature Dominance biases the prediction, leading to inaccurate classifications, as proximity is defined by arbitrary magnitude rather than true statistical similarity.

Technique	Method	Range / Distribution	Sensitivity to Outliers	
Standardization (Z-Score)	Centers data around the mean and scales by the standard deviation	Results in a distribution with mean 0 and standard deviation 1	Less sensitive	 The diagram shows the Z-score formula $Z = \frac{(x - \mu)}{\sigma}$. Red arrows point from the text labels to the corresponding parts of the formula: 'data point' points to x, 'Mean' points to μ, and 'Standard deviation' points to σ.
Normalization (Min-Max)	Rescales feature values to a specific, predefined range, typically	All values are strictly bounded within the defined range	Highly sensitive	 The diagram shows the Min-Max normalization formula $X_{\text{normalized}} = \frac{X - X_{\min}}{X_{\max} - X_{\min}}$. Red arrows point from the text labels to the corresponding parts of the formula: 'Original feature value' points to X, 'Maximum value' points to $X_{\max}$, and 'Minimum value' points to $X_{\min}$.



KNN Overview





ADVANTAGES	DISADVANTAGES
Lazy Learning/Adaptability	Handling Imbalanced Data
Non-Parametric	Sensitivity
Simplicity and Interpretability	Computational Complexity



Real-World Examples



Regression Use Cases

- Mechanism: Predicts a continuous value by calculating the average (or weighted average) of the target values of the K nearest neighbors.
- Example: **House Price Prediction:** To estimate the price of a new house, KNN finds the K closest houses (based on size, age, bedrooms, etc.) and averages their known sale prices to predict the new property's value.¹³

Classification Use Cases

- **Recommendation Systems:** Identifying similar items or users. For example, clustering hotels based on features (ratings, price, location) to provide recommendations for new users based on their nearest neighbors.
- **Marketing and Ad Targeting:** Classifying new customers into specific segments (e.g., demographics, purchasing behavior) to serve them tailored advertisements.